

Stock Analyser App

Product Requirements Document

v3.1 - Production hardening: error logging, query defaults, mobile pass

Updated: 2026-04-30 | Tracks app version: 1.5.0 "Console"

Rename applied: this PDF uses Stock Analyser App as the product name throughout.

This document supersedes PRD v3.0. It captures all features shipped through v1.5.0 "Console", completing the v1.4.x–v1.5.0 hardening series: error boundary (1.4.1), retries+timeouts (1.4.2), a11y + input audit (1.4.3), skeletons + empty states (1.4.4), error logging + query defaults + mobile pass (1.5.0).

1. Release Status

Version	Codename	Date	Theme
1.0.0	Atlas	2026-04-29	Stable baseline: screener, terminal, scores, AI, auth, portfolio, alerts
1.1.0	Beacon	2026-04-29	Landing page split from workspace; signed-in auto-redirect to /app
1.2.0	Compass	2026-04-29	Landing flow restructured into 7-section conversion funnel
1.3.0	Console	2026-04-29	Top nav restructured into 4 grouped dropdowns (Bloomberg/Koyfin pattern)
1.3.5	Console	2026-04-29	Patch: deep-link /terminal/\$symbol no longer hard-404s
1.4.0	Console	2026-04-29	Automated regression test suite (Vitest + Playwright) gated on every build
1.4.1	Console	2026-04-29	App-wide error boundary — render errors never blank-screen
1.4.2	Console	2026-04-29	Patch: retry + timeout on all outbound API calls
1.4.3	Console	2026-04-29	Patch: a11y pass (skip-link, global focus ring) + server-input audit

2. Product Overview

Stock Analyser App is a keyboard-driven, AI-augmented research terminal for global equities. It blends Bloomberg-style data density with an AI co-pilot layer, targeting retail and prosumer investors who want fast research, transparent scoring, and live monitoring across US, India, Europe, Japan, Hong Kong, Korea, Taiwan, Singapore, and Australia.

2.1 Audience

- Self-directed retail investors building conviction before trading.
- Prosumer / power users who want a Bloomberg-like density without the cost.
- Long-term portfolio trackers who want unified holdings + alerts.

2.2 Positioning

"One terminal, three workflows" — value investors, momentum traders, and portfolio trackers each have a tailored entry path through the same data and scoring engine.

3. Information Architecture

As of v1.3.3 "Console", the application top navigation is grouped into 4 dropdown menus. Protected workspace calls now attach the active auth session token explicitly, preventing transient unauthorised responses from reaching the router error boundary.

Group	Items	Purpose
Research	Screener · Analysis · Compare	Discovery and per-ticker deep dive
Workspace	Watchlists · Portfolio · Alerts	Personal, signed-in user state
Market	Events	Cross-market context (earnings, macro)
System	Data Quality · Sources · Settings	Transparency and configuration

Active group highlighting: when any child route is active, its parent group label renders in primary color. Brand collapses to "SAA" abbreviation below sm breakpoint; version tag hidden below md to keep the header single-row.

4. Functional Requirements

4.1 Research

- Screener: filter the global universe by region, sector, valuation (P/E, P/B, dividend yield), momentum (RSI, ROC, MA cross), and proximity to 52-week range. Visualize as table, cards, or sector heatmap.
- Analysis (Terminal): per-ticker page with snapshot, price chart, technicals, fundamentals, transparent Value/Momentum/Quality/Risk/Confidence scores, AI narrative, news catalysts with cited sources, and diff mode for snapshot comparison.
- Compare: side-by-side view of up to 6 tickers with best/worst highlighting per metric.

4.2 Workspace (signed-in)

- Watchlists: user-curated lists of tickers. Three local lists today (My, Value Candidates, Momentum Candidates).
- Portfolio: holdings with live valuation, unrealized P&L, allocation breakdown by sector and region. RLS-protected per user.
- Alerts: rule-based on price, RSI, 52-week range proximity, 5-day momentum. Server-side evaluation with in-app notifications and 12h cooldown per rule.

4.3 Market

- Events: cross-market earnings and macro calendar (placeholder data today — see Section 7 open items).

4.4 System

- Data Quality: per-metric freshness, completeness, and source attribution.
- Sources: transparency panel listing every external feed with status.
- Settings: currency toggle, account, preferences.

4.5 Cross-cutting

- Auth: email + Google OAuth. RLS on portfolio/alerts/user data.
- AI co-pilot: command bar (Cmd/Ctrl-K) for ticker lookup and route navigation.
- Keyboard shortcuts: ? for help, e to export, g+p / g+w for nav.
- Currency toggle (USD/EUR), PDF export, sparklines, diff mode.

5. Non-Functional Requirements

- Stack: TanStack Start v1, React 19, Vite 7, Tailwind v4, Cloudflare Workers (edge SSR).
- Backend: Lovable Cloud (Postgres + Auth + RLS) via server functions.
- AI: Lovable AI Gateway (Gemini / GPT models) for narrative & catalysts.
- Performance: header is single-row sticky, sub-200ms route transitions, preload-on-intent for nav links.
- Accessibility: keyboard-first, focus-visible on all interactive elements, tooltips wired via title attributes on grouped nav triggers.
- SEO: per-route head() metadata; semantic HTML; JSON-LD on root.

6. Design System

- Theme: dark, terminal-style. Primary green/cyan accent, dense mono typography for data, Inter for chrome.
- All colors via semantic tokens in `src/styles.css` (oklch). No raw color classes in components.
- Typography: JetBrains Mono for data and nav labels (uppercase, tracking-wider, 10–11px). Inter for prose.
- Component library: shadcn/ui (Radix primitives) — DropdownMenu, Tooltip, HoverCard, Dialog, Toaster (sonner).

7. Open Roadmap

7.1 Workspace sync

- Cloud-synced watchlists — migrate `use-watchlist.ts` off `localStorage` to a `watchlists` table with RLS.
- Saved screener filter sets — new `saved_screens` table; users can name and reload filter combinations.

7.2 Background alerts

- Scheduled evaluation via `pg_cron` — alerts must fire when no client is open. Today `evaluateMyAlerts` is pull-only.
- Optional email delivery toggle (Resend) — currently in-app toast + bell only.

7.3 Compliance

- `/privacy` and `/terms` routes — prerequisites for public auth signup.

7.4 Data integrity

- Roll out `SourcedValue<T>` across all metrics — per-value source, timestamp, freshness, and confidence.
- "Verified by 2+ sources" badge logic.
- Split `finimpulse.server.ts` into layered providers: `HistoricalPrice`, `Fundamentals`, `Fx`, `SymbolResolver`, `Events`.

7.5 Other

- Real Events calendar data (currently placeholder).
- Unit tests for `src/lib/scores.ts`, `RSI/ROC/MA` calculations, filter logic.
- Backtest / historical replay on screener presets.
- Mobile polish for the terminal table.
- Per-route `og:image` for social sharing.

8. Acceptance Criteria — v1.3.3 "Console"

- Top nav renders exactly 4 group triggers: Research, Workspace, Market, System.
- Each trigger opens a Radix DropdownMenu with the items defined in Section 3.
- Clicking a child link navigates and closes the menu.
- When the active route belongs to a group, the group label renders in primary color.
- Header stays single-row at 1024px and above; brand abbreviates to "SAA" below sm; version tag hidden below md.
- Currency toggle, ? shortcut button, AlertBell, and AuthNav remain right-aligned and unchanged in behavior.
- Each group trigger exposes a title attribute listing its child items (tooltip support).
- Glossary contains a "Nav Groups" entry describing the 4-group structure.

9. Documentation Invariant

Per project memory, every shipped feature/UX change must sync the following on the same release:

- PRD (this document) — version + section affected.
- README — version badge, getting-around section.
- CHANGELOG — Keep-a-Changelog entry under the new version.
- In-app glossary (`src/lib/glossary.ts`) — new or changed concepts.

- Tooltips — title/aria attributes on new chrome.
- Version file (src/lib/version.ts) — APP_VERSION + APP_CODENAME.

v1.3.3 "Console" sync status: version file, README, changelog, PRD, and glossary updated 2026-04-30.

10. Glossary

Term	Definition
Nav Groups	Top nav structure: 4 dropdowns (Research, Workspace, Market, System).
Sub-nav	Optional contextual second row inside route layouts (e.g. ticker tabs in Analysis).
Console	v1.3.0 codename — nav restructure release.
Compass	v1.2.0 codename — 7-section landing flow.
Beacon	v1.1.0 codename — landing/workspace split.
Atlas	v1.0.0 codename — stable baseline.
SourcedValue	Generic wrapper attaching source, timestamp, freshness, and confidence to a metric.
Diff Mode	Compare-page toggle highlighting metric deltas between snapshots.

v2.3 / App v1.3.2 — Popup-blocker resilient auth

When the browser blocks the new auth tab, the app now falls back to in-tab navigation while preserving the user's original location. After successful sign-in (email or Google), a success screen offers two actions: 'Back to where you were' (restores the original URL with the active session) and 'Continue to the terminal' (/app). The popup tab itself gains an explicit 'Cancel and close this tab' control. The original-URL handoff uses sessionStorage key 'auth:return-to'.

v2.4 / App v1.3.3 — Auth transport hardening

Protected alert and portfolio server calls now include the active session bearer token explicitly. AlertBell waits for a valid token before listing, evaluating, or marking alert events, and degrades to an empty notification state on transient auth errors instead of surfacing a raw 401 Response.

/alerts and /portfolio pass auth headers to list, create, update, delete, and evaluate operations. If auth is still settling after popup or same-tab sign-in, both routes show safe empty states rather than a blank-screen router error.

v2.5 / App v1.3.4 — Account menu

Signed-in users now see an account dropdown in the top nav (right of the alert bell) that exposes Switch user... and Sign out. Switch user signs the current user out and immediately opens the auth popup so a different account can sign in without losing the current tab. Sign out ends the session and confirms via toast. The trigger displays the user's email at md+ widths and falls back to a one-letter avatar at narrow widths.

v1.4.4 — Automated Regression Suite

Released 2026-04-30. Adds an automated regression test suite that runs on every build, blocking deploys on failure. Two layers: Vitest unit tests for pure logic (format, scores, backtest) and Playwright headless E2E covering all public routes, the /terminal/AAPL deep-link, the screener filter flow, and auth gating for /portfolio and /alerts.

Coverage

- Public routes: /, /app, /compare, /watchlist, /events, /data-quality, /sources, /changelog, /settings
- Terminal: /terminal/AAPL deep-link mounts and analyzes (validates v1.3.5 fix)
- Auth: /portfolio and /alerts redirect or gate signed-out users; /auth form renders
- Screener: rows render, search filter narrows the universe
- Watchlist: controls visible

Operational notes

CI workflow at .github/workflows/regression.yml runs unit then E2E on every push and PR. Local: bun run test:regression. Authenticated CRUD flows are covered today as gating checks; full-session E2E with a mocked Supabase token is queued as

the next increment.

v1.4.4 — Console (patch) · 2026-04-30

Resilience: App-Wide Error Boundary

A global React error boundary (`<AppErrorBoundary>`) now wraps `<Outlet />` in the root route. Any uncaught render-time or hook-time error in any page or descendant component is captured and rendered as a recoverable card with the error message, a Try again button (resets the boundary), and a Home link — eliminating the blank white screen failure mode.

Coverage

- Router-level `defaultErrorComponent` (existing) catches loader throws.
- `AppErrorBoundary` (new) catches render errors, hook errors, and `useQuery` callback throws.
- Together, every code path that can throw on a route now has a recoverable UI.

Observability Hook

`componentDidCatch` logs the error and component stack to console. This is the designated integration point for Sentry (or equivalent) when production error tracking is enabled.

Files Changed

- `src/components/error-boundary.tsx` (new)
- `src/routes/__root.tsx` (wraps `Outlet`)
- `src/lib/version.ts` (1.4.0 → 1.4.4)
- `CHANGELOG.md`, `README.md` (synced)

v1.4.4 — Console (patch) · 2026-04-30

Resilience: Hardened Outbound HTTP

All outbound HTTP calls now route through `fetchWithRetry`, a shared helper providing exponential backoff (250ms → 1s → 4s, 3 attempts) and per-attempt timeouts (12s default; 30s for AI and news). Retries are scoped to network errors, timeouts, 408, 425, 429, and 5xx. Deterministic 4xx responses bypass retry.

Affected Integrations

- Finimpulse market data (`analyze.ts`, `finimpulse.server.ts`) — universe load + ticker analyze pipelines.
- Lovable AI gateway (`ai.server.ts`) — narrative thesis + K command bar.
- Perplexity news (`news.functions.ts`) — News & Catalysts panel on terminal page.

Operational Effect

A single transient 502/503/504 from any upstream no longer fails the user-facing operation. Each retry is logged with status and label for observability without surfacing to the UI.

Files Changed

- `src/server/http.server.ts` (new)
- `src/server/finimpulse.server.ts`, `src/server/analyze.ts` (use `fetchWithRetry`)
- `src/server/ai.server.ts` (use `fetchWithRetry`, 30s timeout)
- `src/server/news.functions.ts` (use `fetchWithRetry`, 30s timeout)
- `src/lib/version.ts` (1.4.1 → 1.4.4)
- `CHANGELOG.md`, `README.md` (synced)

v1.4.4 — Console (patch) · 2026-04-30

Accessibility & Hardening Pass

Two-part patch: a small but high-impact accessibility pass on the global shell, plus a formal audit of every server function's input validation.

Accessibility Improvements

- Skip-to-main-content link in `__root.tsx` — visible on keyboard focus, jumps past the nav on every route.
- Global `:focus-visible` outline (2px amber ring, 2px offset). Keyboard-only — mouse users unaffected.
- Confirmed every route has a `<main>` landmark and every icon-only button (alert bell, currency toggle, command bar trigger, shortcuts hint, AI co-pilot) carries an `aria-label` or `aria-pressed`.

Server-Input Validation Audit

Reviewed all `createServerFn` entry points. Result: every mutation already uses `Zod.inputValidator()` with appropriate constraints (uuid, regex on tickers, min/max on numerics and strings). No unvalidated server entry points remain. No code changes required.

Files Changed

- `src/routes/__root.tsx` (skip-link + `#main-content` wrapper)
- `src/styles.css` (focus-visible + sr-only utilities)
- `src/lib/version.ts` (1.4.2 → 1.4.4)
- `CHANGELOG.md`, `README.md` (synced)

v1.4.4 — UX Polish: Skeletons & Empty States

Released: 2026-04-30 · Codename: Console · Patch release

Why

Following the resilience work in v1.4.2 (retries + timeouts on external APIs), successful requests can take longer in the worst case. Plain centered “LOADING UNIVERSE...” strings made waits feel longer and zero-data screens (empty watchlist, no holdings, no alerts, no events) terminated with terse one-liners that didn't tell users what to do next. v1.4.4 raises the floor of perceived quality across every data-heavy panel.

What shipped

- Shared feedback primitives in `src/components/feedback-states.tsx`:
 - `EmptyState` — title, optional description, optional icon and action.
 - `EmptyStateLink` — consistent terminal-styled CTA link.
 - `TableSkeleton` — shimmering row placeholders with configurable column count.
 - `StatGridSkeleton` — KPI tile placeholders.
 - `ParagraphSkeleton` — multi-line text placeholders for AI/narrative panels.
- Skeleton loaders applied to:
 - Screener (`/app`) — 9-col × 10-row skeleton table replaces single `LOADING` line.
 - Watchlist (`/watchlist`) — 8-col × 6-row skeleton.
 - Compare (`/compare`) — 6-col × 8-row skeleton.
 - Portfolio (`/portfolio`) — 8-col × 6-row skeleton on positions table.
 - Alerts (`/alerts`) — 6-col × 4-row skeleton on active alerts.
 - Events (`/events`) — 5-col × 6-row skeleton.
 - Data Quality (`/data-quality`) — stat grid + 6-col × 8-row skeleton.
 - AI Narrative panel — 6-line paragraph skeleton during generation.
 - News & Catalysts panel — 6-line paragraph skeleton during Perplexity search.
- Friendlier empty states:
 - Portfolio: icon + 'Add your first holding' CTA.

- Watchlist: explains how to add tickers, links to screener.
- Compare: tells user to pick 2–6 stocks, links to screener.
- Events: separate copy for empty watchlist vs no events in window.
- Alerts: explains alert types and refresh cadence.
- Screener: explains why filters returned zero, with reset CTA.

Acceptance

- First paint of every data-heavy route preserves layout (no content jump on load).
- Every zero-data state renders the shared EmptyState component, never a bare 'No X yet.' string.
- All skeletons carry aria-busy="true" for screen-reader awareness.
- Visual language (font-mono uppercase tracking) consistent across all loading and empty states.

Roadmap status (after 1.4.4)

Hardening remaining: #10 toast consistency audit, #11 keyboard shortcuts expansion, #12 mobile responsiveness pass, #13 PDF export of analysis. Rate-limiting (#3) still deferred until backend primitives improve.

v1.5.0 — Production Hardening

Released: 2026-04-30 · Codename: Console · Minor release

Why

The v1.4.x series made the app safer (boundaries, retries, validation) and more polished (skeletons, empty states). v1.5.0 closes the operational gap: we now see real-world errors instead of guessing, every route uses the same caching/retry policy, and the mobile experience is verified clean at 375px.

What shipped

- Error logging pipeline:
 - New error_logs table (user_id, message, stack, component_stack, route, user_agent, app_version).
 - RLS: users insert own/anonymous logs only; users read own; admin reads all.
 - reportError server function — Zod-validated, payload-capped, attributes via verified bearer token (never trusts body user_id).
 - logClientError client helper — dedupes identical errors within 60s, swallows own failures.
 - Wired into AppErrorBoundary, router defaultErrorComponent, and global unhandledrejection/error listeners.
- Centralised React Query defaults:
 - staleTime 60s, gcTime 5min — match the slowest data refresh cadence.
 - refetchOnWindowFocus off, refetchOnReconnect on — financial data already polls; refocus thrash removed.
 - networkMode online — no fetches when browser reports offline.
 - Smart retry: skip 400/401/403/404/422 (deterministic), retry transient once.
 - Mutations never auto-retry.
- Mobile responsiveness pass:
 - Verified hero, screener, terminal, compare, data-quality, watchlist at 375×812.
 - All wide tables already wrap in overflow-x-auto — confirmed.
 - Added body { overflow-x: hidden } as defense-in-depth.

Acceptance

- Errors thrown anywhere in the app appear in error_logs within seconds.
- Same query key returns cached data across routes for 60s without refetch.
- 401/403/404 never trigger retry storms.
- No page-level horizontal scrollbar at 375px on any tested route.

Roadmap status (after 1.5.0)

Hardening series complete. Remaining polish (optional): #10 toast consistency audit, #11 keyboard shortcut expansion (g h / g p / g w), #13 PDF export of analysis. Rate-limiting (#3) still deferred until backend primitives improve.